

Power in Practice

Full Formal Research Report

Rethinking Electricity Use in Everyday Systems: A Household and Urban Electricity Efficiency Study in Toronto, Ontario

Abstract

Electricity is one of the most important resources in modern urban life. It powers homes, transportation systems, office buildings, communication networks, and digital infrastructure. However, discussions about electricity efficiency often focus mainly on energy production or renewable energy expansion, while less attention is given to how electricity is actually used in daily life. This project investigates inefficient electricity use in household and urban environments through direct observation, Ontario electricity pricing data, public electricity system information, and engineering analysis.

The project began through photography-based observation of Toronto at night. Many office towers, streets, and commercial areas remained brightly illuminated even when visible activity appeared low. This observation led to a broader engineering question: how much electricity waste exists not because of energy shortages, but because systems are not optimized?

To explore this issue, the project examined four levels of electricity use: household behavior, urban lighting conditions, Ontario electricity pricing systems, and grid-level peak demand. A single-household case study was compared with Ontario residential benchmark values. Toronto Hydro pricing plans, including Time-of-Use (TOU) and Ultra-Low Overnight (ULO), were analyzed to understand how price signals influence electricity usage behavior. Ontario's peak demand management framework was also reviewed using Independent Electricity System Operator (IESO) information.

The findings suggest that electricity inefficiency often comes from two repeated causes: daily human behavior and static system design. Small but repeated waste, poorly timed electricity use, and infrastructure that does not adapt to real demand can create avoidable stress on the electricity system. The report concludes that many effective solutions do not require expensive new infrastructure. Instead, meaningful improvements can come from smarter scheduling, adaptive systems, low-cost controls, and better public awareness.

1. Introduction

Electricity demand continues to increase as cities become larger, more digital, and more electrified. In Ontario, population growth, electric vehicle adoption, and building electrification are increasing pressure on the electricity system. At the same time, energy efficiency remains one of the fastest and lowest-cost ways to improve sustainability.

Many discussions about energy focus mainly on supply, such as building new power generation, expanding renewable energy, or improving transmission infrastructure. These topics are important. However, demand-side behavior is also critical because every kilowatt-hour that is unnecessarily consumed must still be generated, transmitted, and paid for.

This project focuses on the following research question:

How can inefficient electricity use in household and urban environments be identified and improved through observation, data analysis, and engineering design?

Unlike purely theoretical energy studies, this project uses real urban observations, Ontario electricity pricing data, and realistic household scenarios to investigate where inefficiency appears in daily life.

2. Project Motivation

The inspiration for this project came from photography.

While photographing Toronto at night, many buildings and commercial districts appeared brightly illuminated even when visible activity was low. Empty office floors remained lit, exterior lighting continued late into the night, and some public spaces appeared brighter than necessary.

This observation became an engineering question rather than only an aesthetic one.

Modern systems already automate transportation, communication, and security systems. However, many lighting and electricity systems still operate in static ways without adapting to actual occupancy or demand.

Because of this, the project transformed urban photography observations into a broader electricity efficiency study focused on real-world systems.

3. Research Questions

Main Research Question

How can inefficient electricity use in household and urban environments be identified and reduced through observation, data analysis, and engineering design?

Sub-Questions

1. How do household behaviors create avoidable electricity waste?
 2. Does urban nighttime lighting exceed practical demand?
 3. How do electricity prices influence user behavior?
 4. How can system design reduce peak demand and unnecessary electricity use?
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4. Methodology

This project used four complementary research methods.

4.1 Direct Observation

Photography and urban observation were used to document lighting conditions in streets, towers, and commercial areas throughout Toronto.

4.2 Household Case Study

A representative household electricity profile was compared with Ontario benchmark usage values to evaluate potential inefficiency patterns.

4.3 Public Data Review

Ontario public electricity pricing and grid information were reviewed, including data and reports from Toronto Hydro, the Ontario Energy Board (OEB), and the Independent Electricity System Operator (IESO).

4.4 Engineering Interpretation

Observed inefficiencies were translated into practical engineering solutions involving automation, scheduling, sensors, controls, and pricing incentives.

5. Household Case Study

5.1 Ontario Residential Benchmark

The Ontario Energy Board states that the typical residential electricity customer benchmark remains approximately **750 kWh per month** (Ontario Energy Board, n.d.).

This benchmark provides a useful reference point for evaluating residential electricity consumption.

Table 1

Monthly Household Usage Scenarios

Household Type	Monthly Use (kWh)	Difference from Benchmark
Efficient Home	620	-17%
Typical Ontario Home	750	Baseline
High-Use Home	900	+20%

5.2 Interpretation

A monthly difference of approximately 130–150 kWh can result from several factors, including:

- HVAC use intensity
- Appliance scheduling
- Standby electricity loads
- Lighting behavior

- Number of occupants
- Building insulation quality

Table 2

Example Daily Waste Patterns

Observation	Waste Type	Potential Fix
Lights on during daylight	Avoidable lighting load	Use natural light
Laundry during peak hours	Timing inefficiency	Shift to off-peak
Chargers constantly plugged in	Standby load	Use smart plugs
Excessive heating/cooling	Control inefficiency	Smart thermostat

5.3 Key Conclusion

Small daily behaviors accumulate over time. However, shifting high-load activities such as laundry, dishwashing, and charging often creates larger savings than only reducing minor lighting loads.

6. Ontario Electricity Pricing Analysis

Ontario allows eligible residential customers to choose between several pricing plans, including Time-of-Use (TOU), Ultra-Low Overnight (ULO), and Tiered pricing (Toronto Hydro, n.d.).

6.1 Toronto Hydro TOU Winter Rates

Table 3

TOU Winter Pricing

Period	Time	Price
Off-Peak	7 PM – 7 AM	9.8¢/kWh
Mid-Peak	11 AM – 5 PM	15.7¢/kWh
On-Peak	7–11 AM, 5–7 PM	20.3¢/kWh

6.2 ULO Rates

Table 4

ULO Pricing

Period	Time	Price
Ultra-Low Overnight	11 PM – 7 AM	3.9¢/kWh
On-Peak	4 PM – 9 PM	39.1¢/kWh

6.3 Interpretation

Ontario's pricing model demonstrates an important engineering principle:

Demand matters as much as total consumption.

The same amount of electricity can create different system impacts depending on when it is used. For example, running a dishwasher at 1:00 AM and 6:00 PM may consume similar electricity, but 6:00 PM usage occurs during a peak demand period.

6.4 Example Load Shift Savings

If 50 kWh of flexible monthly electricity use is shifted from the 39.1¢ ULO peak period to the 3.9¢ overnight period:

Difference = 35.2¢/kWh

Savings:

$50 \times \$0.352 = \17.60 per month

This example shows that scheduling electricity use can significantly reduce electricity costs and peak demand stress.

7. Ontario Grid Context

Ontario's electricity system operates in real time. Electricity supply and demand must remain balanced continuously.

The Independent Electricity System Operator (IESO) tracks electricity demand forecasts and system peaks through Peak Tracker tools (IESO, n.d.).

Ontario is generally considered a summer-peaking province, but winter demand can also increase during evening hours when residents return home, turn on lighting, prepare meals, and use heating systems.

7.1 Why Peak Demand Matters

The electricity grid must be designed to handle the highest-demand hours, not average demand.

As a result, avoidable peak demand can increase:

- Infrastructure costs
- Capacity requirements
- System complexity
- Long-term investment needs

7.2 Engineering Insight

Reducing one kilowatt-hour during peak periods can sometimes be more valuable than reducing one kilowatt-hour overnight because peak demand affects infrastructure planning and grid reliability.

8. Urban Lighting Observations

Although not every illuminated building is inefficient, repeated observations throughout Toronto suggested several common patterns.

8.1 Typical Nighttime Conditions

- Empty office floors remained illuminated
- Decorative façade lighting continued after business hours
- Commercial signage operated in low-activity environments
- Public spaces were lit uniformly regardless of occupancy

8.2 Possible Causes

Several possible explanations include:

- Static timer systems
- Security policies
- Lack of occupancy sensors
- Older infrastructure standards
- Convenience-focused operation

8.3 Limitation of Observation

Observation alone cannot confirm internal occupancy or exact electricity loads. However, observation can still identify areas where adaptive systems and smarter controls may improve efficiency.

9. Key Findings

Finding 1: Household Habits Matter

Repeated behaviors such as unnecessary lighting and standby power create avoidable electricity consumption.

Finding 2: Timing Is Critical

Electricity pricing strongly suggests that when electricity is used can matter almost as much as how much electricity is used.

Finding 3: Peak Demand Is a System Issue

Even moderate residential demand can become costly when concentrated during peak periods.

Finding 4: Urban Systems Are Often Static

Many lighting systems appear designed around fixed schedules instead of adaptive operation based on occupancy or demand.

Finding 5: Low-Cost Controls Offer Strong Value

Timers, sensors, and automated scheduling can create meaningful improvements without requiring major infrastructure investments.

10. Engineering Solutions

10.1 Household Solutions

Table 5

Household Efficiency Solutions

Solution	Cost	Potential Impact
LED replacement	Low	Medium

Smart plugs	Low	Medium
Shift flexible loads	Low	High
Smart thermostat	Medium	High

10.2 Urban Solutions

Table 6

Urban Lighting Solutions

Solution	Cost	Potential Impact
Motion sensors	Medium	Medium
Adaptive brightness	Medium	High
Zoned lighting	Medium	High
Updated schedules	Low	Medium

10.3 Grid-Level Solutions

Table 7

Grid-Level Solutions

Solution	Cost	Potential Impact
Dynamic pricing	Low	High
Demand response	Medium	High
Smart EV charging	Medium	High
Real-time dashboards	Medium	Medium

11. Discussion

This project suggests that electricity inefficiency is rarely caused by one major failure. More often, it results from many small decisions repeated over time.

Examples include:

- Leaving systems on because adjusting them is inconvenient
- Using electricity during expensive peak periods out of habit
- Designing infrastructure around worst-case assumptions instead of real demand patterns

From an engineering perspective, this means the opportunity is not only technological innovation, but also optimization.

Modern cities already possess large electricity infrastructure systems. The challenge is operating these systems more intelligently and efficiently.

12. Limitations

This project has several limitations.

First, the household analysis used benchmark comparisons rather than direct smart meter data. Second, urban lighting observations were qualitative rather than based on measured lux values. Third, household electricity use can vary significantly depending on heating type, occupancy, and building characteristics. Finally, some illuminated buildings may have legitimate operational or security-related reasons for remaining lit.

These limitations do not remove the value of the project, but they define its scope as an exploratory engineering study.

13. Future Work

Future versions of this project could include:

- Real smart meter household data
 - Light intensity sensor measurements
 - Comparisons across multiple Toronto neighborhoods
 - Predictive electricity demand models
 - Interactive dashboards connected to Ontario live data
 - AI-based scheduling recommendations for households
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14. Conclusion

Electricity efficiency is often discussed in abstract terms, but it is visible in everyday systems.

It appears in hallway lights left on all day.

It appears in dishwashers running during peak hours.

It appears in office towers glowing late into the night.

It appears in infrastructure designed around peak demand rather than optimized usage.

This project combined observation, public data, and engineering reasoning to study these patterns.

The strongest conclusion is that meaningful efficiency improvements do not always require expensive new infrastructure. Many improvements can come from:

- Better timing
- Better controls
- Better incentives

- Better system design

For future cities such as Toronto, smarter electricity use may become just as important as producing more electricity.

References

[Independent Electricity System Operator \(IESO\)](#)

[Ontario Energy Board \(OEB\)](#)

[Toronto Hydro Residential Rates](#)